



AI IN BUSINESS

A technical foundation to understanding how AI systems work, and how to best leverage the technology.

Event Resources:





PRESENTERS

About Us



Tom von den Steinen

Information Systems & Data Science

Deloitte.



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Information Systems & Computer Science

KPMG



The Technology Consulting Workshop

- Helps students explore tech consulting and real-world applications through intimate networking opportunities with industry professionals.
- Prepares members for internships and careers in consulting and tech-driven industry roles based on projects and group cases.

Important Dates (Fall Semester)

— TCW Fall Callout Meeting (TOMORROW)

- Tuesday, September 29th CG0036, 7-8 PM
- Dress Code: Casual

— TCW x Accenture Case Competition

- **Signups:** September 29th – October 15th
- **Kickoff:** Thursday, October 15th HH2083, 6:30-7:30 PM
- **Presentation Day:** Friday, October 23rd

— Fall Coffee Chats

- Friday, November 6th & Sunday, November 8th
- HH 4103 from 2-5 PM, Group Format



Jason Tang

VP of Outreach

jptang@iu.edu



Annual Chicago Trip



TCW @ Hoosiers Outrun Cancer

Our Placements



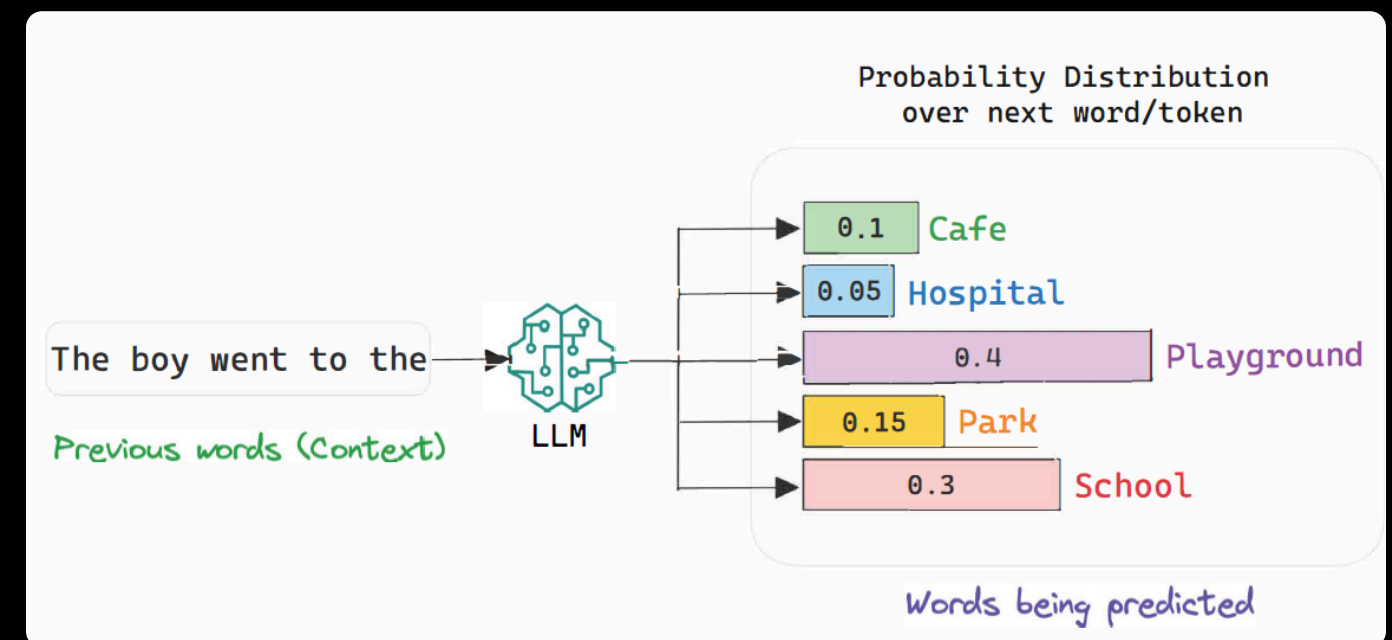


LLMs

LLMs are random word generators

Your job is to guide that randomness

- LLMs predict the next word. They are random word generators that are shaped by their training data and **the context you provide or they autonomously access**.
- LLMs will likely always have pitfalls such as:
 - Hallucinations
 - Knowledge cutoffs & training data quality
 - Limited context window / attention
 - Broad scope, undefined processes
 - Probabilistic and random
- Recent Shifts:
 - Agents gather their own context through tools (web search, your databases, calendar, email).
 - Models run longer and reason more, so exact wording of a prompt matters less than the infrastructure around it.





LLMS

How do we address these pitfalls?

By building systems around LLMs to allow them to **autonomously gather and appropriately manage context**.

What matters now?

Context engineering

- Understanding the components of LLM systems to customize prebuilt AI providers to curate the correct context for a given task/goal.

THEN

Prompt engineering

One crafted prompt



LLM



Output

NOW

Context engineering

Prompt

Tools

Memory

Files

Skills



LLM



Relevant output

How do LLMs build context?



CONNECTORS & TOOLS

Tools

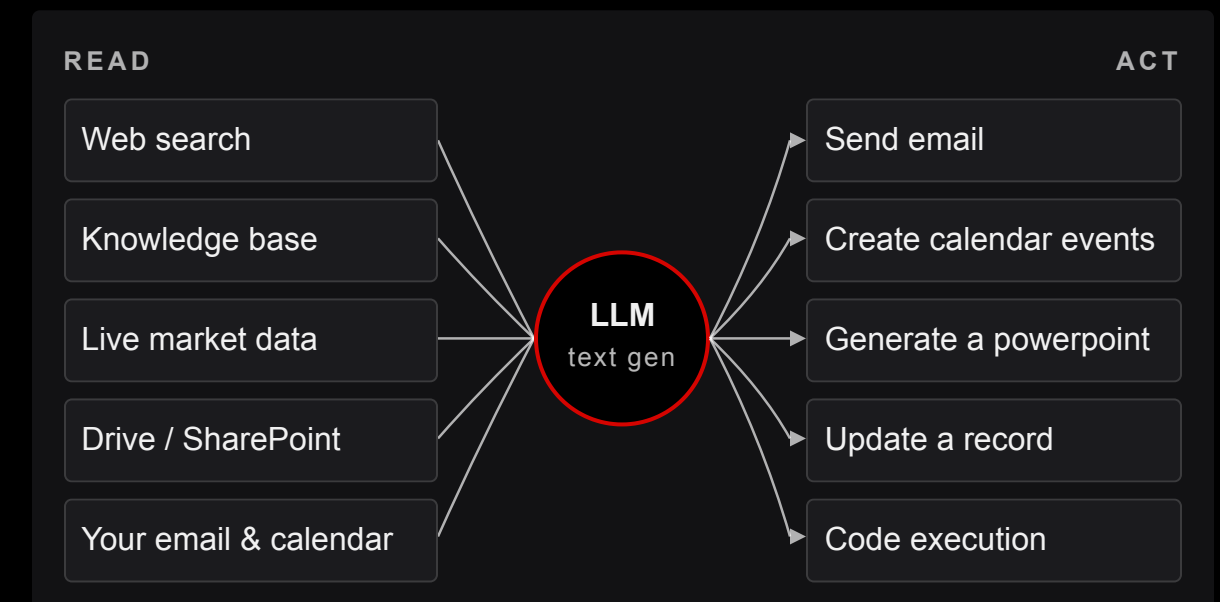
Tools extend the LLM beyond text generation to fetching live information and taking action.

— Understand what the model can actually "see" or access.

- Tools dictate the context that an AI autonomously gathers. Understand what tools you use to understand what information is shaping your LLM's response, and understand when to ask it to use which.
- For example:
 - Slide generation runs through code, so the AI never sees the slide. Quality check means converting slides to images first, then analyzing those.
 - Asking ChatGPT for a YouTube summary is a bad idea -- it has no YouTube tool and will hallucinate. Ask Gemini and you get a factual response, because it has the programmatic ability to fetch information from YouTube.

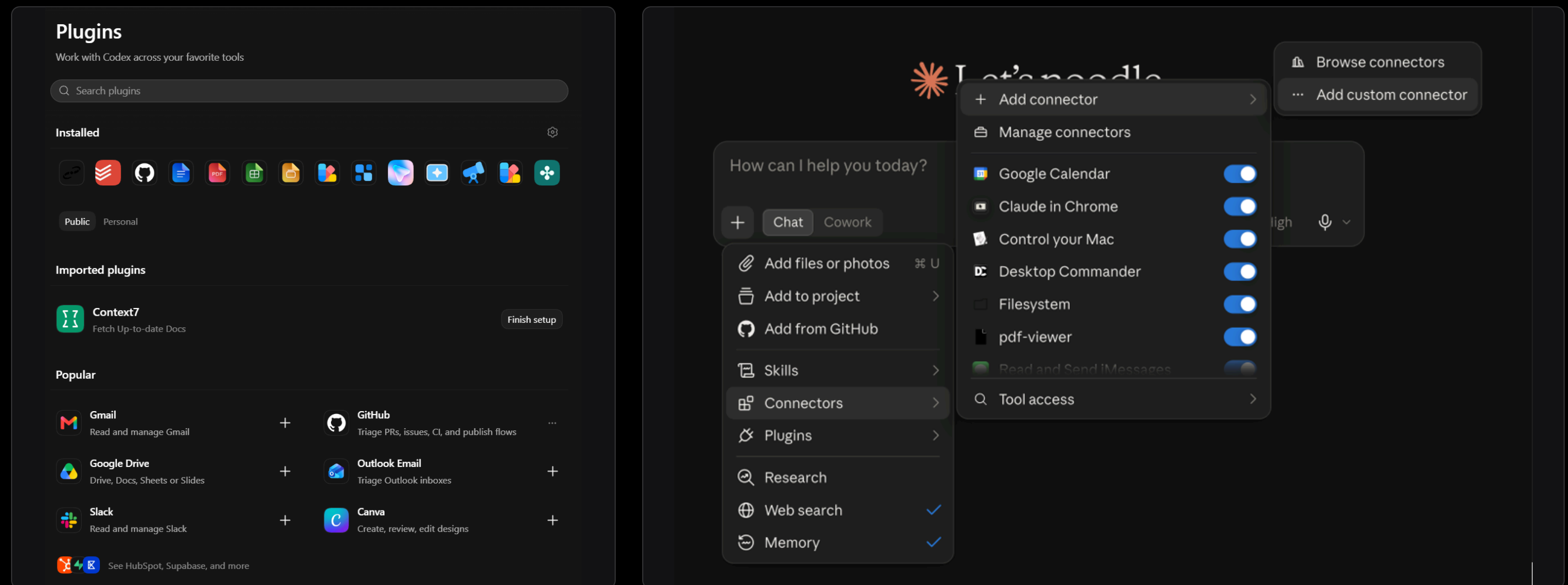
— You can't expect capability you didn't set up.

- AI will hallucinate what it has access to -- don't trust it blindly. Understand its access and capabilities.
- Minor exception: code execution -- AI can increasingly use code to discover tools to find capabilities, but the underlying code/tool must still exist.



CONNECTORS & TOOLS

Tools in AI interfaces





CONNECTORS & TOOLS

Tool use in a chat

Upcoming calendar events ▾

Read my calendar, what do i have coming up

Loaded tools, used Google Calendar integration ▾

Searched available tools **Google Calendar list events** >

Lists calendar events in a given calendar. >

Here's what's coming up over the next two weeks (skipping recurring workout/wake-up routines):

Today (Sept 11)

- Coding Exercise 1 (Week 3) due
- **TCW Chicago Trip** starts — 8:00 AM to 10:45 PM
- Visit to PwC — 11:00 AM-12:30 PM, PwC office, 1 N Wacker Dr, Chicago

Sept 12 — TCW Chi Trip Day 2 & travel back, 10:00 AM-3:00 PM

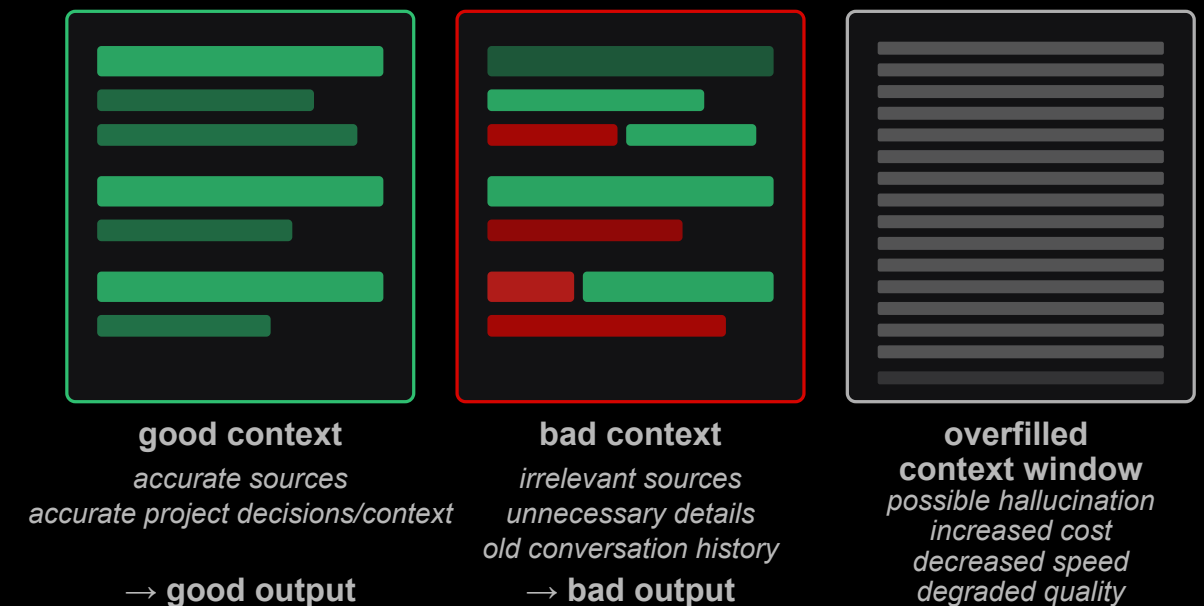
CONTEXT WINDOWS

Context windows

The biggest gap between desired outputs and what was received is often missing context that wasn't mentioned.

- Everything in the context window shapes the LLM's generation.
 - Shape responses through good context via files, prompts, and good information from connectors/tools.
 - Bad context (ex: irrelevant or incorrect web searches) can throw off and pollute the quality of a response.
- Too much context degrades performance.
 - LLMs weigh provided context equally. Stale context (like long conversation history) will impact response generation and increase the chance of hallucinations.
 - Start new chats often. Carry forward only the context that matters.
- In addition to limiting bad/stale context, **provide as much good context as possible**:
 - Brain dumps, voice input, and metaprompting
- **Maintain clear and accurate sources of truth** to quickly provide abundant correct context without retyping it every time.
 - Especially beneficial when completing teamwork with AI, to ensure teammates' AI interactions do not drift apart due to the AI making decisions in ambiguity.

CONTEXT TYPES





SKILLS

Skills: codify repeated prompts, create specific guidelines

A skill is a reusable, natural language prompt that is automatically loaded when needed. Use it where specific instructions and processes benefit output quality, and to save time on manual prompting.

- Use for:
 - Repeatable workflows, processes, or guidelines.
 - Protocols or preferences on how to behave when dealing with certain concepts, topics, actions, tools, coding languages, file types, etc.
 - Granular control over process and quality.
- Invest the time to outline the process once. Then, skills will trigger automatically when requested in natural language.
- **Examples:** Word / PPT / Excel generation · Company Analysis · RFP response workflow · Company Branding/Style · Brainstorming Process · Daily Planning · Inbox Triage · Deliverable Generation Principles · Research Processes.

```
---
name: structured-research
description: Runs a disciplined research pass – background,
clarifying questions, targeted search, ranked sources,
hallucination check. Use when the user asks for research,
a policy scan, or a sourced writeup.
---
only name + description sit in context until triggered

# Structured Research Skill

1. Gather background context on the topic
2. Ask clarifying questions to guide the research
3. Targeted search – .gov sources, specific
databases, etc.
You choose the sources, not the model
4. Rank results by relevance to your query
5. Double-check for hallucinations
6. Identify future areas of investigation
or open questions
```

EXAMPLE INPUT

"Use the structured research skill to research policy changes to X"



SKILLS

Adding a skill

Plugins

Manage plugins, skills, and MCPs

Plugins 13

Apps 5

MCPs 3

Skills 1

Technology Consulting Cases

Use when a user wants a technology consulting practice case, wants to facilitate one for a...

Personal ☒

Browse directory

Add

Create plugin

Add a marketplace

Add MCP server

Record a skill

Customize

Skills

Connectors

Plugins

Yours

Discover

Search

Filter

Sort

Add

From Anthropic

Productivity

Tasks, planning and follow-ups — the everyday toolkit.

Add

New

See all

Maven Bio

Evidence-backed BioPharma research workflows over Maven Bio's drug, trial,...

from Maven Bio · added 2 days ago

NetSuite AI Companion

Guides AI assistants that use the NetSuite AI Connector: tool-selection order, outpu...

from Oracle NetSuite · added 2 days ago

NetSuite Finance Analyst

Director-level finance analysis on live NetSuite data: financial reporting, period...

from Oracle NetSuite · added 2 days ago

learn

Use this skill when the user wants intellectual understanding — learning ho...

from Anthropic · added May 28

Unstructured Foundation

Connect sources and search processed documents with Unstructured Foundatio...

added 2 days ago

doc-coauthoring

Guide users through a structured workflow for co-authoring documentation. Use whe...

from Anthropic · added Dec 4, 2025



SKILLS

Using a skill in a chat

Friday 8:20 AM

technology-consulting-cases

Skill Installer install this skill

Worked for 46s >

Installed technology-consulting-cases successfully, including all three reference files. It will be available starting with your next turn.

Technology Consulting Cases help me run an ERP case for my casing partner

Worked for 1m 18s >

ERP Case: Summit Trail Equipment

Facilitator note — do not read aloud

- **Duration:** 25–35 minutes
- **Candidate level:** Undergraduate or early-career
- **Topic:** ERP strategy and implementation
- **Intended solution:** A phased ERP implementation with standardized core processes
- **Evaluates:** Problem diagnosis, stakeholder thinking, scope decisions, data readiness, testing, rollout, change management, and executive communication
- **Format:** Seven stages plus a final recommendation
- **No calculations required**

Use optional probes selectively. The candidate does not need to mention every answer-guide item.

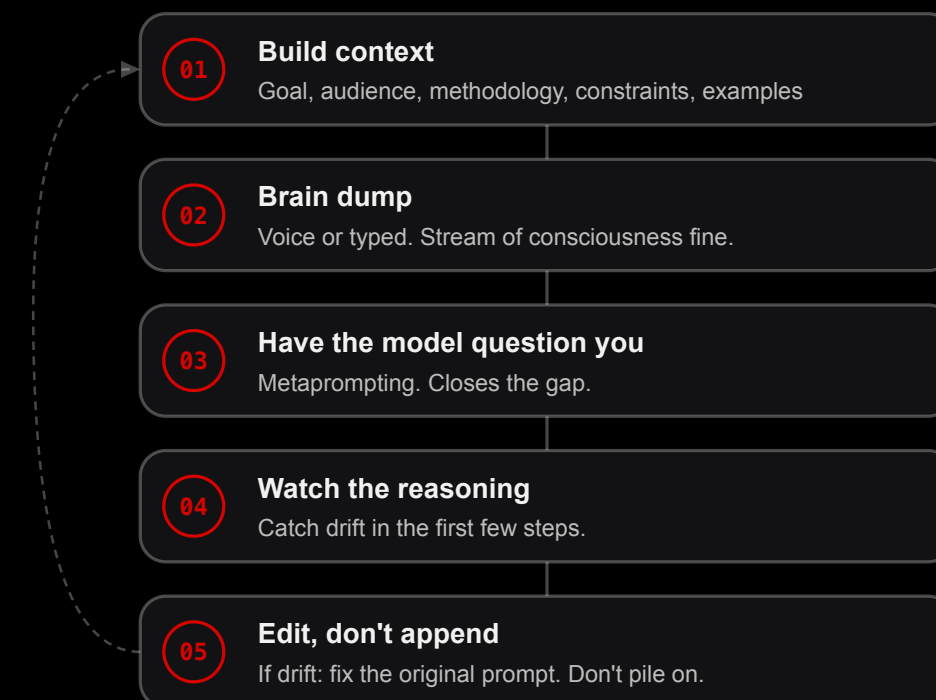


CONTEXT ENGINEERING

Prompting Framework

Curate abundant good context. Leaving gaps creates room for the LLM to make incorrect assumptions and drift from your intent.

- Guide the model's reasoning and audit what enters the context window -- skills, tool calls, web results (quality and recency) -- especially at the start of a run.
 - Bad or stale context early sends the whole task in the wrong direction. Catching drift in reasoning saves time versus waiting for a finished output.
 - Hallucinations often start in early reasoning -- "I don't have access to that information, let me guess." If something looks off, stop and edit your initial prompt rather than stacking follow-ups.
- Editing the original prompt reduces context usage -- improving response quality, mitigating hallucinations, and reducing cost.
- Metaprompting surfaces things for you to clarify that you hadn't previously thought of, and lets the model write its own prompts.



What makes an agent?

How does LLM technology get leveraged into agents that take action?



AGENTS

Components of an agent

The agent harness provides these components to the LLM technology in a loop.

Harness / environment

Prompt

Goal, audience, methodology, constraints, examples.

System prompt

Standing instructions sent with every message.

Tools / connectors

Read from and act on the outside world.

Memory

What persists across sessions.

Files

Source material and deliverables that persist across sessions.

Code execution

Compute and build files rather than just describe them.

Skills

Instructions loaded only when they are needed.

Agent Loop

Provides a durable loop between chat requests that allows agents to work toward multi-step goals



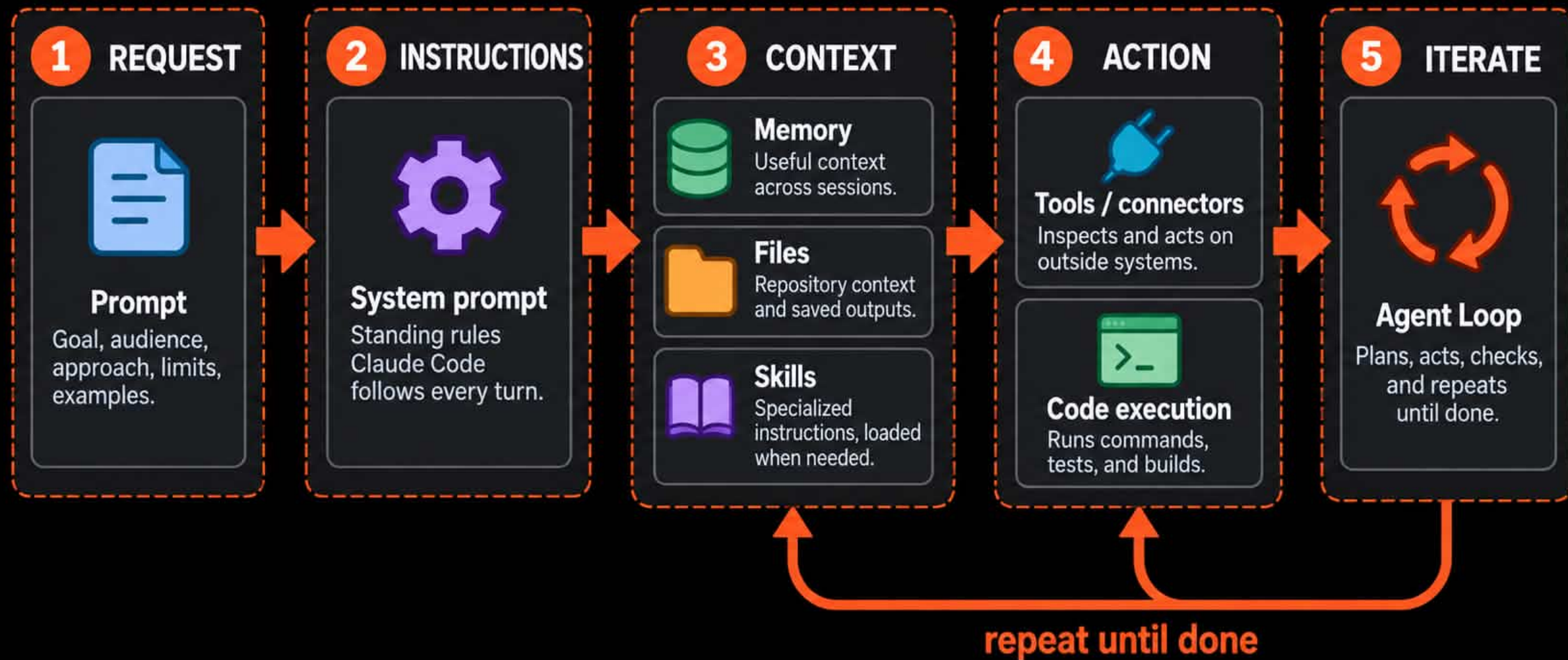
LLM

Text Generation



CLAUDE CODE HARNESS

How a coding request moves through the harness





TOOL SELECTION

Types of General Agent Environments

Where the model runs determines what capabilities it has and how well you can leverage the underlying LLM.

REGULAR · PROJECTS

Chat window

- Chats are for one-off tasks. **Projects** are chats with pre-loaded context -- files and custom instructions injected into every new chat.
- Best for an **individual use case** or **repetitive action**, when the underlying files and instructions stay the same.
- Stable context, low setup. Good when you don't need the AI wandering a folder or building files on its own.
- **System instructions** set rules sent with every message (e.g. always reference [file] before responding).

Best for: stable, repeatable individual workflows, or quick questions and one off tasks.

CLAUDE COWORK / CHATGPT WORK

Folder agent

- Operates in a **folder**. Agentic -- it navigates the folder to build context as needed, and creates files automatically.
- Deliverables **persist**: PowerPoints, models and docs stay there across sessions, and context carries forward.
- **Best for iterating** when context evolves as you work. Terminal access means more capability, more risk.
- **System instructions**: a `claude.md` or `agents.md` file is read with every message.

Best for: iterating, multi-file knowledge work.

CODEx AND CLAUDE CODE

Coding agent

- More applicable to technical work than knowledge work.
- Full coding focus. The agent can install software and use the terminal.
- Same folder + terminal idea as Cowork/Codex, but optimized for building real tools.
- Most power, most risk -- essentially gives the AI its own computer.

Best for: building, dev work and expanded computer, terminal, and tool access.

SIMPLER · SAFER



MORE CAPABLE · MORE RISK



MODEL SELECTION

Model Selection

Model size supplies capability. Thinking level supplies deliberation.

Reasoning

offmax

The amount of time a model "thinks" before generating a final response. Improves accuracy, improves instruction following, and decreases hallucinations.

Model size

SML

The underlying power of the model. Larger means more training data and more capable. Performance is still dependent on reasoning time.

SIZE	CLAUDE	GEMINI	CHATGPT
Small	Haiku	Flash-Lite	Luna
Medium	Sonnet	Flash	Sol
Large	Opus	Pro	Astra



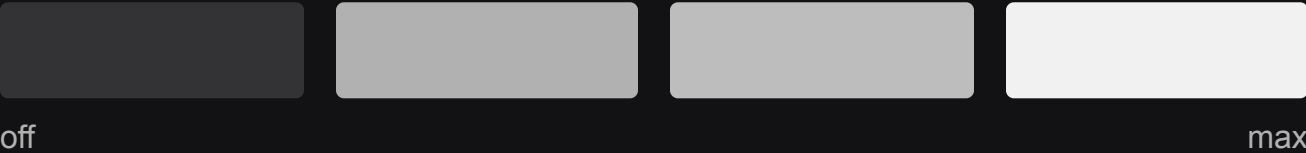
MODEL SELECTION

Model Selection

Model size supplies capability. Thinking level supplies deliberation.

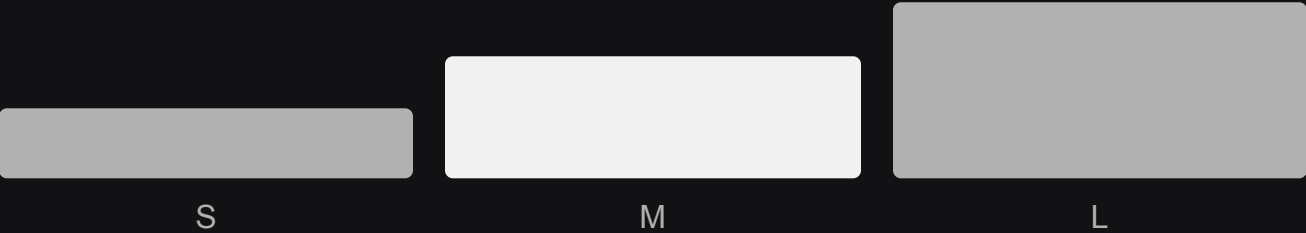
Bigger models don't fix bad context. The best models with max reasoning still have the same inherent pitfalls of LLMs (ex: limited training data cutoff, and potential for bad search results). **Steer the context.**

Reasoning



The amount of time a model "thinks" before generating a final response. Improves accuracy, improves instruction following, and decreases hallucinations.

Model size



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Medium	Sonnet	Flash	Sol
Large	Opus	Pro	Astra

MODEL SELECTION

Choosing by Scenario

FAST

Simple + easy to verify

- Rename variables in one file
- Reformat notes into bullets
- Answer from supplied text

MODEL

Small

THINKING

Low

DEFAULT

Routine professional work

- Draft a client-ready email
- Add a well-scoped feature
- Summarize several documents

MODEL

Medium

THINKING

Medium

DEEP

Ambiguous or multi-step

- Debug an issue across many files
- Compare architecture options
- Reconcile conflicting evidence

MODEL

Medium

THINKING

High

CRITICAL

High stakes + hard to check

- Final client recommendation
- Security or financial review
- Novel, long-horizon build

MODEL

Large

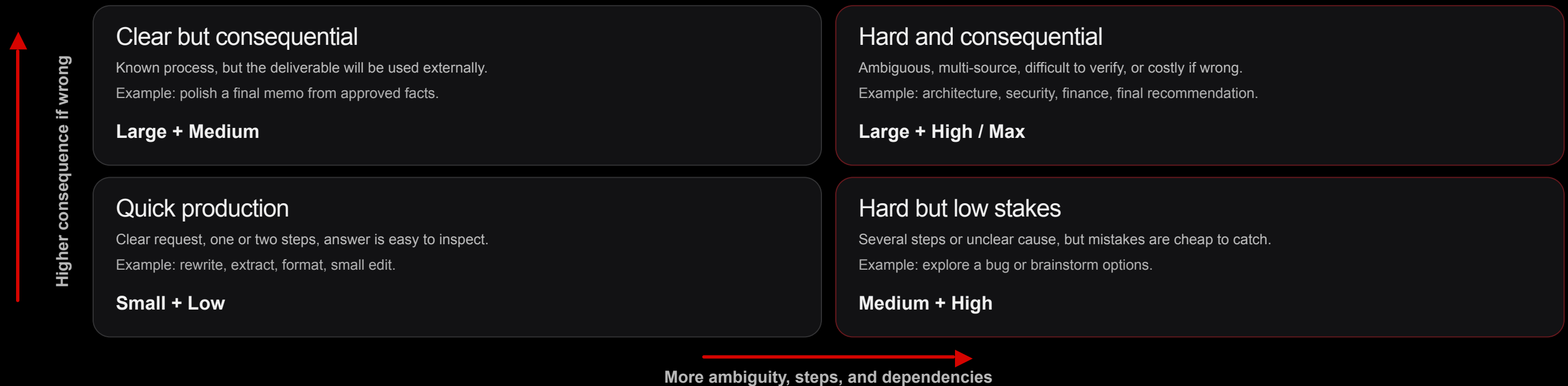
THINKING

High / Max



MODEL SELECTION

Decision Matrix





VERTICAL TOOLS

Vertical AI tools

The aforementioned AI tools attempt to be general purpose. These have a narrow purpose, and are usually much better at the specific task they're intended for.



Perplexity
Web search



NotebookLM
Source-grounded research



Cursor / Windsurf / Devin
Coding agents



Harvey AI
Legal services



Claude Design
Design



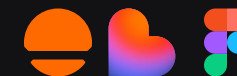
Gamma
Presentations



N8N
Workflow automation



Replit
App development



Base44 / Lovable / Figma
App design & development



STUDENT ACCESS

Free resources and student accounts

TOOL	DESCRIPTION
ChatGPT Pro	Access increased usage limits and premium models with your IU login. Students can also redeem four months of Pro on a personal account: chatgpt.com/students/2026
Gemini Pro	Access increased usage limits and premium models with your IU login
Claude Pro	No Pro access with student email. Anthropic-sponsored clubs sometimes give out premium access (ex: Claude Builder Club)
GitHub Student Developer Pack	Access to GitHub Copilot coding models



Demo Workflow

01

Research

Research Topic

02

Choose direction

Turn research into a slide outline

03

Choose a look

Three visual slide previews

04

Build Slides

HTML deck



Demo Workflow

01

Research

Research Topic

Skill:

topic-research.md

How to research

Output:

research.md

Findings and sources

02

Choose direction

Turn research into a slide outline

Skill:

slide-direction.md

Align with user intent

Output:

slide-brief.md

Approved outline

03

Choose a look

Three visual slide previews

Skill:

html-slides.md

How to design and build slides

Output:

design-comparison.html

Three visual slide previews

04

Build Slides

HTML deck

Skill:

html-slides.md

How to design and build slides

Output:

slides.html

HTML presentation



Demo Workflow

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Research Topic

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`design-comparison.html`

Three visual slide previews

04

Build Slides

HTML deck

Skill:

`html-slides.md`

How to design and build slides

Output:

`slides.html`

HTML presentation

Customize it:

`iu-branding.md`

`making-skills.md`

Slide quality review



Event Resources:



AI IN BUSINESS



Tom von den Steinen

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Joeb Rahman

[LinkedIn](#)